

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

|                      |                          |
|----------------------|--------------------------|
| Product Name         | <u>Toluene</u>           |
| CAS No               | 108-88-3                 |
| Synonyms             | Tol; Methylbenzene       |
| Molecular Formula    | C7 H8                    |
| Molecular Weight     | 92.14                    |
| Recommended Use      | Laboratory chemicals.    |
| Uses advised against | No Information available |

|                         |                                                                                             |
|-------------------------|---------------------------------------------------------------------------------------------|
| Product Code            | <b>C17685</b>                                                                               |
| Address                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| Emergency Tel.          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| Telephone / Fax Numbers | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| E-mail address          | <a href="mailto:ANZinfo@thermofisher.com">ANZinfo@thermofisher.com</a>                      |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

HSNO Approval Number     **HSR001227**

### GHS Classification

#### Physical hazards

Flammable liquids

Category 2

#### Health hazards

|                                                      |            |
|------------------------------------------------------|------------|
| Aspiration Toxicity                                  | Category 1 |
| Acute Oral Toxicity                                  | Category 4 |
| Acute Inhalation Toxicity - Vapors                   | Category 4 |
| Skin Corrosion/Irritation                            | Category 2 |
| Serious Eye Damage/Eye Irritation                    | Category 2 |
| Reproductive Toxicity                                | Category 2 |
| Specific target organ toxicity - (single exposure)   | Category 3 |
| Specific target organ toxicity - (repeated exposure) | Category 2 |

#### Environmental hazards

Chronic aquatic toxicity

Category 3

**Label Elements****Signal Word****Danger****Hazard Statements**

H225 - Highly flammable liquid and vapor  
H304 - May be fatal if swallowed and enters airways  
H315 - Causes skin irritation  
H336 - May cause drowsiness or dizziness  
H361 - Suspected of damaging fertility or the unborn child  
H373 - May cause damage to organs through prolonged or repeated exposure  
H412 - Harmful to aquatic life with long lasting effects  
H319 - Causes serious eye irritation  
H302 + H332 - Harmful if swallowed or if inhaled

**Precautionary Statements****Prevention**

P202 - Do not handle until all safety precautions have been read and understood  
P201 - Obtain special instructions before use  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P233 - Keep container tightly closed  
P242 - Use non-sparking tools  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P243 - Take action to prevent static discharges  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear eye protection/ face protection

**Response**

P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P308 + P313 - IF exposed or concerned: Get medical advice/attention  
P331 - Do NOT induce vomiting  
P332 + P313 - If skin irritation occurs: Get medical advice/attention  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

Toxic to terrestrial vertebrates

## Section 3 - Composition and Information on Ingredients

| Component | CAS No   | Weight % |
|-----------|----------|----------|
| Toluene   | 108-88-3 | <=100    |

## Section 4 - First Aid Measures

### Description of first aid measures

|                                            |                                                                                                                                                                                                     |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                                                                                              |
| <b>New Zealand Emergency Tel.</b>          | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                                                                                                                          |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur. Risk of serious damage to the lungs (by aspiration).                                   |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                     |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                                                                   |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water. Do NOT induce vomiting. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward. |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                    |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Causes central nervous system depression: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting              |
| <b>Notes to Physician</b>                  | Treat symptomatically. Smallest quantities reaching the lungs through swallowing or subsequent vomiting may result in lung edema or pneumonia. Symptoms may be delayed.                             |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

Do not use water jetstream.

### Specific Hazards Arising from the Chemical

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

**Personal Precautions, Protective Equipment and Emergency Procedures****Emergency procedures**

Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

**Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling****Advice on safe handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Ensure adequate ventilation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

**Conditions for Safe Storage, Including any Incompatibilities****Storage Conditions**

Keep containers tightly closed in a dry, cool and well-ventilated place. Flammables area. Keep away from heat, sparks and flame.

**Incompatible Materials**

Strong oxidizing agents. Strong acids. Strong bases. Halogenated compounds.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

**Control parameters****Exposure limits**

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

| Component | New Zealand WEL                          | Australia                                    | ACGIH TLV   | The United Kingdom                                         |
|-----------|------------------------------------------|----------------------------------------------|-------------|------------------------------------------------------------|
| Toluene   | TWA: 20 ppm<br>TWA: 75 mg/m <sup>3</sup> | STEL: 150 ppm<br>STEL: 574 mg/m <sup>3</sup> | TWA: 20 ppm | STEL: 100 ppm 15 min<br>STEL: 384 mg/m <sup>3</sup> 15 min |

|  |                                                      |                                           |  |                                                             |
|--|------------------------------------------------------|-------------------------------------------|--|-------------------------------------------------------------|
|  | STEL: 100 ppm<br>STEL: 377 mg/m <sup>3</sup><br>Skin | TWA: 50 ppm<br>TWA: 191 mg/m <sup>3</sup> |  | TWA: 50 ppm 8 hr<br>TWA: 191 mg/m <sup>3</sup> 8 hr<br>Skin |
|--|------------------------------------------------------|-------------------------------------------|--|-------------------------------------------------------------|

**Biological limit values**

**NZ** - Substances assigned Biological Exposure Indices in the New Zealand Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**ACGIH** - American Conference of Governmental Industrial Hygienists (ACGIH) TLVs® and BEIs®- Threshold Limit Values for Chemical Substances and Physical Agents & Biological Exposure Indices. 2022 Edition

| Component | New Zealand                                                                                                                           | Australia | ACGIH - Biological Exposure Indices                                                                                                                                                                                                                                            | United Kingdom |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Toluene   | 0.03 mg/L (urine) end of exposure or end of shift (Toluene)<br>0.3 mg/g creatinine (urine) end of exposure or end of shift (O-Cresol) |           | 0.02 mg/L<br>Medium: blood<br>Time: prior to last shift of workweek<br>Determinant: Toluene<br>0.03 mg/L<br>Medium: urine<br>Time: end of shift<br>Determinant: Toluene<br>0.3 mg/g creatinine<br>Medium: urine<br>Time: end of shift<br>Determinant: o-Cresol with hydrolysis |                |

**Appropriate engineering controls****Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Individual protection measures, such as personal protective equipment**

**Eye Protection** Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments                                                                                                            |
|----------------|-------------------|-----------------|-----------------|---------------------------------------------------------------------------------------------------------------------------|
| Viton (R).     | < 240 minutes     | 0.30 mm         | AS/NZS 2161     | Permeation rate 68 µg/cm <sup>2</sup> /min As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
| Viton (R)      | > 480 minutes     | 0.70 mm         |                 |                                                                                                                           |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)

**Recommended half mask:-** Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

|                                        |                                                                                                 |
|----------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                          |
| <b>Environmental exposure controls</b> | Prevent product from entering drains. Do not allow material to contaminate ground water system. |

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                                              |                                          |
|------------------------------------------------|----------------------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                                       |                                          |
| <b>Appearance</b>                              | Colorless                                    |                                          |
| <b>Odor</b>                                    | aromatic                                     |                                          |
| <b>Odor Threshold</b>                          | 1.74 ppm                                     |                                          |
| <b>pH</b>                                      | No information available                     |                                          |
| <b>Melting Point/Range</b>                     | -95 °C / -139 °F                             |                                          |
| <b>Softening Point</b>                         | No data available                            |                                          |
| <b>Boiling Point/Range</b>                     | 111 °C / 231.8 °F                            | @ 760 mmHg                               |
| <b>Flammability (liquid)</b>                   | Highly flammable                             | On basis of test data                    |
| <b>Flammability (solid,gas)</b>                | Not applicable                               | Liquid                                   |
| <b>Explosion Limits</b>                        | <b>Lower</b> 1.2 vol%<br><b>Upper</b> 7 vol% |                                          |
| <b>Flash Point</b>                             | 4 °C / 39.2 °F                               | <b>Method</b> - No information available |
| <b>Autoignition Temperature</b>                | 535 °C / 995 °F                              |                                          |
| <b>Decomposition Temperature</b>               | No data available                            |                                          |
| <b>Viscosity</b>                               | 0.6 mPa.s @ 20 °C                            |                                          |
| <b>Water Solubility</b>                        | practically insoluble 0.5 g/L @ 20°C         |                                          |
| <b>Solubility in other solvents</b>            | No information available                     |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                              |                                          |
| <b>Component</b>                               | <b>log Pow</b>                               |                                          |
| Toluene                                        | 2.73                                         |                                          |
| <b>Vapor Pressure</b>                          | 29 mbar @ 20 °C                              |                                          |
| <b>Density / Specific Gravity</b>              | 0.866                                        |                                          |
| <b>Bulk Density</b>                            | Not applicable                               | Liquid                                   |
| <b>Vapor Density</b>                           | 3.1                                          | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)                      |                                          |

### Other information

|                             |                                                           |
|-----------------------------|-----------------------------------------------------------|
| <b>Molecular Formula</b>    | C7 H8                                                     |
| <b>Molecular Weight</b>     | 92.14                                                     |
| <b>Explosive Properties</b> | Not explosive Vapors may form explosive mixtures with air |
| <b>Oxidizing Properties</b> | Not oxidising                                             |
| <b>Evaporation Rate</b>     | 2.4 (Butyl acetate = 1.0)                                 |

## Section 10 - Stability and Reactivity

|                                         |                                            |
|-----------------------------------------|--------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available |
| <b>Stability</b>                        | Stable under normal conditions.            |
| <b>Sensitivity to Mechanical Impact</b> | No information available                   |
| <b>Sensitivity to Static Discharge</b>  | No information available                   |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.   |
| <b>Hazardous Reactions</b>              | None under normal processing.              |

**Conditions to Avoid** Incompatible products, Excess heat, Keep away from open flames, hot surfaces and sources of ignition.

**Incompatible Materials** Strong oxidizing agents, Strong acids, Strong bases, Halogenated compounds.

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## Section 11 - Toxicological Information

### Acute Effects

### Information on likely routes of exposure

#### Product Information

**Inhalation** Irritating to respiratory system. May be harmful if inhaled. May cause drowsiness and dizziness.

**Eyes** Irritating to eyes.

**Skin** Irritating to skin. Can be absorbed through skin. May be harmful in contact with skin.

**Ingestion** Aspiration hazard if swallowed - can enter lungs and cause damage. May be harmful if swallowed. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea. Harmful if swallowed. Potential for aspiration if swallowed.

### Numerical measures of toxicity

#### (a) acute toxicity;

**Oral** Based on available data, the classification criteria are not met

**Dermal** Based on available data, the classification criteria are not met

**Inhalation** Based on available data, the classification criteria are not met

| Component | LD50 Oral            | LD50 Dermal            | LC50 Inhalation       |
|-----------|----------------------|------------------------|-----------------------|
| Toluene   | > 5000 mg/kg ( Rat ) | 12000 mg/kg ( Rabbit ) | 26700 ppm ( Rat ) 1 h |

#### (b) skin corrosion/irritation;

**Test method** Category 2  
OECD 404

**Test species** rabbit

**Observational endpoint** Irritating to skin

(c) serious eye damage/irritation; Based on available data, the classification criteria are not met

#### (d) respiratory or skin sensitization;

**Respiratory** Based on available data, the classification criteria are not met

**Skin** Based on available data, the classification criteria are not met

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

Not mutagenic in AMES Test

(f) carcinogenicity; Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

#### (g) reproductive toxicity;

**Reproductive Effects** Category 2  
Experiments have shown reproductive toxicity effects on laboratory animals

**Developmental Effects** Developmental effects have occurred in experimental animals

**Teratogenicity** Possible risk of harm to the unborn child

(h) STOT-single exposure; Category 3

Results / Target organs Central nervous system (CNS)

(i) STOT-repeated exposure; Category 2

Target Organs Liver, Kidney, Central nervous system (CNS), Blood, spleen, Neuropsychological effects, Eyes, Ears.

(j) aspiration hazard; Category 1

#### Symptoms / effects, both acute and delayed

Causes central nervous system depression. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

## Section 12 - Ecological Information

### Ecotoxicity

#### Aquatic ecotoxicity

The product contains following substances which are hazardous for the environment. Contains a substance which is: Toxic to aquatic organisms.

| Component | Freshwater Fish                                                                                              | Water Flea                                                                                         | Freshwater Algae                                                                                                                 | Microtox                |
|-----------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Toluene   | 50-70 mg/L LC50 96 h<br>5-7 mg/L LC50 96 h<br>15-19 mg/L LC50 96 h<br>28 mg/L LC50 96 h<br>12 mg/L LC50 96 h | EC50: = 11.5 mg/L, 48h<br>(Daphnia magna)<br>EC50: 5.46 - 9.83 mg/L,<br>48h Static (Daphnia magna) | EC50: = 12.5 mg/L, 72h static<br>(Pseudokirchneriella subcapitata)<br>EC50: > 433 mg/L, 96h<br>(Pseudokirchneriella subcapitata) | EC50 = 19.7 mg/L 30 min |

Terrestrial ecotoxicity There is no data for this product

Persistence and Degradability Readily biodegradable

Persistence Persistence is unlikely.

| Component                     | Degradability |
|-------------------------------|---------------|
| Toluene<br>108-88-3 ( <=100 ) | 86% (20d)     |

Degradation in sewage treatment plant Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

Bioaccumulative Potential Bioaccumulation is unlikely

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Toluene   | 2.73    | 90                            |

Mobility The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Spillage unlikely to penetrate soil. The product is insoluble and floats on water. Is not likely mobile in the environment due its low water solubility.

### Other adverse effects

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors  
Persistent Organic Pollutant This product does not contain any known or suspected substance  
Ozone Depletion Potential This product does not contain any known or suspected substance



## Section 13 - Disposal Considerations

### Waste treatment methods

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.                                                                                    |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                                                                                                                                |
| <b>Other Information</b>                   | Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations . Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not let this chemical enter the environment. Do not empty into drains. |

## Section 14 - Transport Information

| Component                     | Hazchem Code |
|-------------------------------|--------------|
| Toluene<br>108-88-3 ( <=100 ) | 3YE          |

### NZS 5433:2020

|                             |         |
|-----------------------------|---------|
| <b>UN-No</b>                | UN1294  |
| <b>Proper Shipping Name</b> | TOLUENE |
| <b>Hazard Class</b>         | 3       |
| <b>Packing Group</b>        | II      |

### IATA

|                             |         |
|-----------------------------|---------|
| <b>UN-No</b>                | UN1294  |
| <b>Proper Shipping Name</b> | TOLUENE |
| <b>Hazard Class</b>         | 3       |
| <b>Packing Group</b>        | II      |

### IMDG/IMO

|                             |         |
|-----------------------------|---------|
| <b>UN-No</b>                | UN1294  |
| <b>Proper Shipping Name</b> | TOLUENE |
| <b>Hazard Class</b>         | 3       |
| <b>Packing Group</b>        | II      |

|                                                                                 |                                                                                                                         |
|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Environmental hazards</b>                                                    | No hazards identified                                                                                                   |
| <b>Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code</b> | Not applicable, packaged goods                                                                                          |
| <b>Special Precautions</b>                                                      | No special precautions required. Please refer to the applicable dangerous goods regulations for additional information. |
| <b>Additional information</b>                                                   | None known                                                                                                              |

## Section 15 - Regulatory Information

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

|                             |           |
|-----------------------------|-----------|
| <b>HSNO Approval Number</b> | HSR001227 |
|-----------------------------|-----------|

**National Regulations**

Any applicable tolerable exposure limits and environmental exposure limits according to the EPA Controls for Hazardous Substances are listed below

| Component | Tolerable Exposure Limit (TEL) Air | Tolerable Exposure Limit (TEL) Water | Tolerable Exposure Limit (TEL) Surface | Environmental Exposure Limits (EEL) |
|-----------|------------------------------------|--------------------------------------|----------------------------------------|-------------------------------------|
| Toluene   | 400 µg/m <sup>3</sup>              | 0.8 mg/L                             |                                        | 330 µg/L (Water)                    |

**Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Authorisation/Restrictions according to EU REACH**

| Component | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Toluene   | -                                                                   | Use restricted. See item 48.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

**International Inventories**

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component | CAS No   | NZIoC | AICS | EINECS    | ELINCS | NLP | KECL     | IECSC | TCSI |
|-----------|----------|-------|------|-----------|--------|-----|----------|-------|------|
| Toluene   | 108-88-3 | X     | X    | 203-625-9 | -      | -   | KE-33936 | X     | X    |

| Component | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | PICCS | ISHL | ENCS |
|-----------|----------|------|-----------------------------------------------|-----|------|-------|------|------|
| Toluene   | 108-88-3 | X    | ACTIVE                                        | X   | -    | X     | X    | X    |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**Section 16 - Other Information**

**This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations**

#### Legend

|                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                    | <b>AICS</b> - Australian Inventory of Chemical Substances                                                                    |
| <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                      | <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances |
| <b>DSL/NDL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                      | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                                                  |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                     | <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                            | <b>CAS</b> - Chemical Abstracts Service                                                                                      |
| <b>TWA</b> - Time Weighted Average                                                                   | <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists                                                     |
| <b>IARC</b> - International Agency for Research on Cancer                                            | <b>PNEC</b> - Predicted No Effect Concentration                                                                              |
| <b>NZS 5433:2020</b> - Transport of Dangerous Goods on Land                                          | <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code                            |
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships                  | <b>ADG</b> - Australian Code for the Transport of Dangerous Goods by Road and Rail                                           |
| <b>LD50</b> - Lethal Dose 50%                                                                        | <b>LC50</b> - Lethal Concentration 50%                                                                                       |
| <b>EC50</b> - Effective Concentration 50%                                                            | <b>ATE</b> - Acute Toxicity Estimate                                                                                         |
| <b>WEL</b> - Workplace Exposure Limit                                                                | <b>RPE</b> - Respiratory Protective Equipment                                                                                |
| <b>DNEL</b> - Derived No Effect Level                                                                | <b>NOEC</b> - No Observed Effect Concentration                                                                               |
| <b>POW</b> - Partition coefficient Octanol:Water                                                     | <b>BCF</b> - Bioconcentration factor                                                                                         |
| <b>vPvB</b> - very Persistent, very Bioaccumulative                                                  | <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              |
| <b>VOC</b> - (Volatile Organic Compound)                                                             |                                                                                                                              |

#### **Key literature references and sources for data**

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).  
<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS  
EPA Guide to classifying hazardous substances in New Zealand  
EPA - Assigning a product to an existing HSNO approval guide

#### **Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.  
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.  
First aid for chemical exposure, including the use of eye wash and safety showers.  
Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.  
Chemical incident response training.

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|-------------------------|-----------------|
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#### **Disclaimer**

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## End of Safety Data Sheet